Answer RFIs before they become delays

RFI Copilot for Summit Ridge Builders · Engineering & Construction · General contractor, 8 active and recent projects, $995M contract value

About $1.6M a year in protected margin, roughly half a point of operating margin. RFI response time drops from 7.9 to 6 days. RFIs likely to miss their SLA are flagged the day they are submitted, before the delay reaches the critical path and the bonus scorecard.

Built end to end on the Databricks Data Intelligence Platform. All figures come from the working prototype running on synthetic but realistic portfolio data.

Why we are here: the industry is losing margin to late information

What the industry data shows	Source
Large projects typically finish **20% late** and run **up to 80% over budget**	McKinsey Global Institute, *Reinventing Construction* (2017)
Construction productivity grew about **1% a year** for two decades, versus 2.8% for the world economy	McKinsey Global Institute (2017)
Project teams spend **35% of their time** (over 14 hours a week) searching for information, resolving conflicts and fixing rework	FMI & PlanGrid, *Construction Disconnected* (2018)
Bad data cost the industry an estimated **$1.8 trillion** in 2020, and about **14% of rework** traces to bad data	Autodesk & FMI, *Harnessing the Data Advantage* (2021)
The average project generates **796 RFIs**, each taking **9.7 days** to answer and costing about **$1,080** to process	Navigant Construction Forum (2013)

What we see in Summit Ridge's own data: 47% of RFIs miss the design team's SLA, 44 of the 65 open RFIs are already late, and late critical-path RFIs carry **$11.2M** of liquidated-damages exposure.

The RFI is where late information turns into lost margin. It is the moment a crew stops and waits.

Why invest now, and why build it on your own data

- **Contracts have shifted schedule risk onto the GC.** Harbor Point carries $60K/day in liquidated damages, so schedule certainty now decides both margin and the next bid.
- **AI can now read the contract documents.** In the prototype, retrieval finds the governing spec section for 95% of open RFIs and Claude drafts a cited answer, at pay-per-token cost.
- **The data already exists, just disconnected.** RFI logs, spec books, firm SLAs and schedules sit in Procore or ACC exports and PDFs. Unifying them once in a governed lakehouse also feeds the next use cases: submittals, change orders and schedule risk.
- **Building it beats buying a point tool.** Off-the-shelf RFI tools stop at the workflow. This keeps the data, the risk model, the AI and the audit trail inside the GC's own tenant, with no per-seat lock-in and the GC's history as the training set.
- **Waiting has a price.** Each quarter of the status quo forgoes about **$0.4M** of the $1.6M a year. Harbor Point is in its peak RFI phase right now (24 open, 15 already late).

The outcome, in your KPIs

KPI (portfolio)	Today	Target with RFI Copilot	Who feels it
Average days to answer an RFI	**7.9 days** (p90: 15 days)	**≤ 6.0 days**	Superintendents, crews
RFIs that miss the contractual SLA	**46.6%**	**≤ 30%**	Project Executives
Liquidated-damages exposure from late critical-path RFIs	**$11.2M** since 2023 (≈ $3.2M/yr)	**–30%** (≈ $1.0M/yr avoided)	COO, CFO
Change-order cost originating in RFIs	**$16.8M** (740 COs)	Root-cause feedback to preconstruction	Preconstruction, Estimating
Project-engineer time per RFI	≈ $1,080 of handling cost per RFI (industry)	–35% with triage + cited AI drafts	Project Engineers

Value: about $1.6M per year, which is roughly **+0.5 points of operating margin** on this portfolio (slide 11).

What this does to the numbers each executive is paid on

Executive	Typically measured and paid on (confirm in discovery)	What RFI Copilot moves	Impact in their terms
COO / VP Operations	Portfolio operating margin vs plan, on-time completion, repeat-owner work	LD days avoided, faster answers, early escalation of high-risk RFIs	**≈ +0.5 pts of margin** on ≈ $285M annual volume, about **+14% net profit** at a 4% net margin
CFO	Net margin vs forecast (no quarter-end fade), forecast accuracy, cash from approved change orders	PCOs flagged at the RFI; schedule risk visible weeks earlier	$16.8M of RFI-driven change orders priced and submitted while the GC still has leverage
Project Executive (Harbor Point)	Project fee and gross-profit fade, milestone dates, owner satisfaction	Day-one risk ranking; approver sign-off on high-stakes answers	Each critical-path week avoided = **$420K of LDs, about 3% of a 5% fee** on a $310M job
Director of Project Controls	EAC forecast accuracy, change-order cycle time, reporting effort	Governed RFI facts, Genie answers, complete audit trail	A weekly risk review on live data instead of spreadsheets

Margin fade and missed milestones are the two things that cut these executives' annual incentive. RFI Copilot attacks both at the RFI, where a large share of the fade begins.

The problem: RFIs are a hidden schedule risk

- **3,795 RFIs** across 8 projects. **65 are open right now, and 44 of them are already past SLA.**
- Half of all RFIs miss the design team's contractual response window (7 days, or 3 days if urgent).
- When a late RFI sits on a **critical-path** activity, every day it slips becomes a liquidated-damages day: $25K/day at Lakeside Hospital and $60K/day at Harbor Point.
- Project engineers triage by gut feel and re-type answers that already exist in the spec book. Leadership has no portfolio view of which firm, discipline or document gap is driving the cost.

Where it hurts most (from the gold tables and Genie)

Project	RFIs	Open / past SLA	Avg days	SLA breach	CO cost	LD exposure
Harbor Point Data Center Ph 2	778	24 / 15	10.5	66%	$3.2M	**$7.6M**
Lakeside Regional Hospital	846	8 / 4	8.5	53%	$4.2M	**$2.6M**
Union Station Mixed-Use Tower	537	6 / 5	7.3	42%	$2.2M	$0.8M

- **Design firms:** the MEP and fire-protection engineers breach SLA 57–65% of the time, versus 21–36% for civil and structural.
- **Root cause (Gen AI on every RFI):** spec ambiguity is 28% of volume, and field conditions plus design conflicts drive **$10.3M** of change-order cost.
- **Implication:** two projects and two design firms account for most of the exposure, and those two Project Executives carry most of the bonus risk. That is where to start.

What changes for the project team (domain owner)

Today: an RFI arrives by email or in Procore. The PE reads it, searches the spec PDF, types an answer, forwards it to the architect and chases it in a spreadsheet.

With RFI Copilot:

1. **One inbox ranked by risk.** Every open RFI shows the ML probability that the design team will miss the SLA, the expected days to answer, and *why* (for example, "MEP firm breaches more often · 26 RFIs on this project in 14 days · no drawing attached").
2. **Triage gate.** If the spec already answers it, the PE closes it in one click and it never reaches the design team. About 28% of RFIs are spec-ambiguity questions.
3. **A cited draft is waiting.** Vector Search finds the governing spec section, and Claude drafts a response that cites it (for example, "per 21 13 13"). The PE edits and approves it.
4. **Approver sign-off (human in the loop).** High-risk, urgent or cost/schedule-impacting responses stop in an Approvers queue for a named Project Executive, who signs off or returns them with a comment. Low-risk answers go straight to the design firm.
5. **Route, flag a PCO, close.** Every action shows a live trace (Lakebase commit in about 5 ms, then the audit job merging into Delta) and lands in the RFI's audit trail.
6. **Ask Genie.** "Which open RFIs on Harbor Point will miss SLA?" returns a plain-English answer plus the SQL behind it.

How it works: one integrated data journey

1. **Lakeflow:** Auto Loader ingests the nightly PM-system exports (RFI logs as JSON, design-firm SLAs as CSV, the spec book as Markdown) into bronze, silver and gold tables. Expectations removed 49 defective or duplicate records in the latest run.
2. **Unity Catalog:** tags, descriptions, a column mask that hides change-order dollars from anyone outside finance, least-privilege grants to the app, and automatic lineage (19 table-to-table edges captured).
3. **ML:** an SLA-breach classifier and a days-to-answer model, tracked in MLflow, registered in Unity Catalog as `@champion`, and run nightly to score every open RFI.
4. **Gen AI:** `ai_classify` assigns a root cause to every RFI. A Vector Search index over the spec book plus Claude (Foundation Model API) produces cited drafts.
5. **Lakebase:** a synced table serves the inbox to the app in about 6 ms. App decisions are Postgres transactions that a Lakeflow Job merges back into a governed Delta audit table.
6. **Genie + Databricks App:** a Genie space over the gold, ML, root-cause and audit tables, and the RFI Copilot app for the project team.

Everything is one Asset Bundle: `databricks bundle deploy && databricks bundle run rfi_build`.

Proof it works (prototype results)

Capability	Result from the latest end-to-end run
Lakeflow data quality	3,844 raw records → 3,795 clean: 15 unknown-project and 15 invalid-discipline rows dropped by expectations, 19 duplicate re-sends removed
ML: who will miss SLA	AUC **0.74** on a time-based holdout; **77% precision in the top-20% risk band vs a 47% base rate (1.64× lift)**
ML: days to answer	MAE 3.5 days vs 4.1 days for a naive baseline
Gen AI retrieval	The governing spec section was retrieved for **95%** of open RFIs (hit@4)
Gen AI drafts	65 of 65 drafts generated; 64 of 65 cite only spec sections present in the retrieved context
Gen AI root cause	`ai_classify` agrees with the ground-truth label for 74% of 3,795 RFIs
Lakebase serving	Point reads **p50 5.8 ms / p95 8.1 ms**; app writes commit in about 5 ms
Genie	6 of 6 executive questions answered with generated SQL (overdue by project, firm speed, design-conflict CO cost, LD exposure)

Value model (annualized), translated into margin

Lever	Mechanism	Assumption	Annual value
Avoided liquidated damages	Escalate high-risk critical-path RFIs at submission, days before the SLA lapses	30% of the ≈ $3.2M/yr LD exposure from late critical-path RFIs avoided	**≈ $0.96M**
Project-engineer productivity	Triage gate + cited drafts	≈ 1,085 RFIs/yr × $1,080 handling cost × 35% saved	**≈ $0.41M**
Fewer RFI-driven change orders	Root-cause feedback to preconstruction (design conflicts + missing information)	10% of ≈ $1.9M/yr	**≈ $0.19M**
Total			**≈ $1.6M / yr**
In margin terms	≈ $285M annual construction volume ($995M over ≈ 3.5 years)	4% net margin ≈ $11.4M profit/yr	**≈ +0.5 pts margin, ≈ +14% profit**

Annual values are history total ÷ 3.5 years. Every assumption, including volume and net margin, is a parameter to validate in discovery and the pilot.

For the executive sponsor (COO / VP Operations)

- **Protects the number on your scorecard.** About half a point of portfolio operating margin, from LD days that never happen and engineering hours that go back into the work, tracked per project in governed tables.
- **No quarter-end surprises.** High-risk RFIs are visible the day they are submitted, not when the fade shows up in the monthly forecast. Genie answers "where are we exposed?" without waiting on an analyst.
- **Leverage with design firms.** Firm-level SLA performance (one specialty engineer breaches 65% of the time) gives you facts for fee negotiations and the next selection.
- **Delivery reputation wins work.** On-time milestones on LD-heavy jobs like Harbor Point are what earn repeat owners and the next negotiated contract.
- **Low-risk platform choice.** One governed platform for data, ML, Gen AI, the app database and BI, running on the GC's existing Databricks account.

For the domain owner (Director of Project Controls / Project Executives)

- **Defends your project's fee.** A week of critical-path slip at Harbor Point costs $420K, about 3% of the fee. The risk-ranked inbox makes sure those RFIs are chased first.
- **Forecasts you can stand behind.** RFI-driven PCOs are flagged at the RFI, so the EAC carries them weeks before close-out.
- **Less typing, same control.** AI drafts are always reviewed and approved by a person. Citations show exactly which spec section supports the answer, and gaps are flagged instead of guessed.
- **Fewer RFIs reach the design team.** The triage gate closes the ones the contract documents already answer.
- **Audit trail by default.** Every approve, route, PCO flag or close is a timestamped event in Lakebase, mirrored to a governed Delta table and queryable in Genie.
- **Fits existing tools.** Ingestion reads the PM system's exports (Procore and Autodesk Construction Cloud style), so no new data entry is required.

Trust, governance and controls

- **Unity Catalog** governs every table, model, index and the synced Lakebase table: lineage from raw file to app, tags, and descriptions.
- **Least privilege:** the app runs as its own service principal with SELECT on specific tables, CAN_USE on one warehouse, CAN_RUN on the Genie space, and INSERT on one Postgres table.
- **Sensitive dollars are masked:** change-order cost is NULL for anyone outside the finance group, enforced by a column-mask function. The build's own run shows the masked output.
- **Human in the loop:** AI drafts never go out without a person approving them. A policy gate (ML risk, AI-assessed impact, urgency, LD rate) adds a named approver for high-stakes answers, and sign-off turnaround is tracked. The decision log is append-only and synced to Delta.
- **No customer data:** the prototype uses synthetic data; a pilot would run inside the GC's own workspace.

90-day pilot, measured on the executive scorecard

Weeks	Scope	Success criteria
1–3	Connect Procore/ACC RFI exports and the spec book for **Harbor Point** and **Lakeside**	Daily Lakeflow ingest; gold KPIs match the PM system within 1%
4–6	Risk model retrained on the GC's own history; Copilot rolled out to 6 project engineers	Top-20% risk band precision ≥ 70%; ≥ 60% of drafts approved with light edits
7–10	Genie space for project executives; weekly risk review	High-risk RFIs escalated ≥ 10 days before the SLA; executives use Genie weekly
11–13	Measure against the scorecard	Avg response ≤ 6 days; **zero LD days from late RFIs on pilot projects; fee-fade trend flat or improving**; go/no-go for all 8 projects

The ask: a sponsor (COO), a domain owner (Director of Project Controls), two pilot projects, read access to RFI exports and spec books, and **the current incentive scorecard metrics, so the pilot reports in exactly those terms.**

Appendix: assumptions and sources

- Portfolio, RFIs, firms and spec book are **synthetic**, produced by `01_generate_raw_data`. The generator encodes realistic drivers: firm responsiveness, coordination load, missing drawings, priority, and a 12% critical-path rate.
- Liquidated-damages exposure = schedule-impact days × the project's LD rate, counted **only** for critical-path RFIs.
- Margin translation: annual volume ≈ $995M ÷ 3.5 years ≈ $285M; a 4% net margin is an illustrative assumption. Harbor Point fee is assumed at 5% of $310M (≈ $15.5M). Executive incentive metrics are typical of GC compensation plans and **must be confirmed in discovery**.
- Industry sources: McKinsey Global Institute, *Reinventing Construction: A Route to Higher Productivity* (2017); FMI & PlanGrid, *Construction Disconnected* (2018); Autodesk & FMI, *Harnessing the Data Advantage in Construction* (2021); Navigant Construction Forum, *Impact of Requests for Information on Construction Projects* (2013).
- Platform run cost (indicative, to be sized in the pilot): a nightly serverless pipeline and jobs, one small Lakebase instance, one Vector Search endpoint, a serverless SQL warehouse, and pay-per-token model calls.
- Code, executed notebooks with outputs, and app smoke-test results: see the GitHub repository (`evidence/`).